

Neighboring readouts, conditioning, and the scope of a slope barrier

Section 3 source note 5 — 25 September 2026

Abstract

For an equal-slope tanh dictionary on a uniform lattice, neighboring differences have whole-line Riesz bounds independent of the number of features after a common spatial scale is removed. Physical tanh readouts on the same zero-sum function space incur an additional condition-number factor proportional to width. This improvement leaves exponential small-slope attenuation intact. We prove these statements, identify a distinct bias-halo obstruction on a finite observation window, and record the exact gradient metric induced by invertible readout coordinates. These results supply controls for interpreting a slope-dependent optimization barrier; they do not establish a universal training lower bound or an error floor.

1 Dictionary, norm, and the comparison being made

Fix a center spacing $h > 0$ and a dimensionless slope $\lambda > 0$. The physical slope is $\gamma = \lambda/h$. For $j = 0, \dots, m$, define

$$\phi_j(x) = \tanh\left(\lambda\left(\frac{x}{h} - j\right)\right), \quad \psi_j = \phi_j - \phi_{j+1} \quad (0 \leq j < m).$$

The neighboring-readout map is

$$B_m : \mathbb{R}^m \longrightarrow L^2(\mathbb{R}, dx), \quad B_m q = \sum_{j=0}^{m-1} q_j \psi_j.$$

Coefficients have the Euclidean norm. Singular values refer to this coefficient norm and the stated function norm; their ratio is denoted κ . Thus a Gram matrix has condition number κ^2 . Using $dx/2$ instead of dx multiplies all squared singular values by 1/2 and leaves κ unchanged.

An individual ϕ_j is not in $L^2(\mathbb{R})$, because it has nonzero tails. The physical readout comparison must therefore use

$$\mathcal{W}_m = \left\{ w \in \mathbb{R}^{m+1} : \sum_{j=0}^m w_j = 0 \right\}, \quad P_m w = \sum_{j=0}^m w_j \phi_j.$$

Every such combination has exponentially decaying tails. The maps P_m and B_m represent exactly the same functions, with different coefficient norms. Neither includes a bias or an unconstrained final tanh anchor. Those columns require a finite observation domain and a separate analysis below.

2 A width-independent whole-line bound

Theorem 1 (Explicit Riesz bounds). *For every $m \geq 1$, $h > 0$, $\lambda > 0$, and $q \in \mathbb{R}^m$,*

$$ha_\lambda \|q\|_2^2 \leq \|B_m q\|_{L^2(\mathbb{R})}^2 \leq 4h \|q\|_2^2, \quad a_\lambda = \frac{4\pi^2}{\lambda^2 \sinh^2(\pi^2/(2\lambda))}. \quad (1)$$

Consequently,

$$\sqrt{ha_\lambda} \leq \sigma_{\min}(B_m) \leq \sigma_{\max}(B_m) \leq 2\sqrt{h}, \quad \kappa(B_m) \leq C_\lambda := \frac{\lambda}{\pi} \sinh\left(\frac{\pi^2}{2\lambda}\right). \quad (2)$$

In particular, the bounds for $B_m/\sqrt{h}$ do not depend on width.

Proof. Use the dimensionless coordinate $t = x/h$. Introduce the probability density and the piecewise constant coefficient function

$$\rho_\lambda(t) = \frac{\lambda}{2} \operatorname{sech}^2(\lambda t), \quad p_q(t) = \sum_{j=0}^{m-1} q_j \mathbf{1}_{[j, j+1]}(t).$$

Integration of the tanh derivative gives the exact identity

$$(B_m q)(ht) = 2(\rho_\lambda * p_q)(t). \quad (3)$$

Since $\|\rho_\lambda\|_1 = 1$ and $\|p_q\|_2^2 = \|q\|_2^2$, the L^1 – L^2 convolution inequality proves the upper bound in (1).

Use the Fourier convention $\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$. The transform of the smoothing density is

$$\hat{\rho}_\lambda(\omega) = M_\lambda(\omega) = \frac{\pi\omega/(2\lambda)}{\sinh(\pi\omega/(2\lambda))}, \quad M_\lambda(0) = 1. \quad (4)$$

For completeness, setting $u = e^{2\lambda t}$ and $a = \omega/(2\lambda)$ reduces the transform to

$$\int_0^\infty \frac{u^{-ia}}{(1+u)^2} du = \Gamma(1-ia)\Gamma(1+ia) = \frac{\pi a}{\sinh(\pi a)}.$$

The first equality is the beta integral and the second is the gamma reflection identity, continued at $a = 0$.

Writing $\operatorname{sinc} u = \sin(u)/u$, with $\operatorname{sinc} 0 = 1$, we also have

$$\hat{p}_q(\omega) = e^{-i\omega/2} \operatorname{sinc}(\omega/2) Q(\omega), \quad Q(\omega) = \sum_{j=0}^{m-1} q_j e^{-ij\omega}.$$

For $|\omega| \leq \pi$, $\operatorname{sinc}(\omega/2) \geq 2/\pi$ and $M_\lambda(\omega) \geq M_\lambda(\pi)$: the function $v/\sinh v$ decreases for $v \geq 0$. Plancherel's identity, restriction to this frequency interval, and Parseval's identity for Q give

$$\begin{aligned} \frac{1}{h} \|B_m q\|_2^2 &= \frac{4}{2\pi} \int_{\mathbb{R}} M_\lambda(\omega)^2 \operatorname{sinc}^2(\omega/2) |Q(\omega)|^2 d\omega \\ &\geq \frac{16}{\pi^2} M_\lambda(\pi)^2 \frac{1}{2\pi} \int_{-\pi}^{\pi} |Q(\omega)|^2 d\omega = a_\lambda \|q\|_2^2. \end{aligned}$$

Taking square roots and ratios proves (2). $\square$

The word ‘‘Riesz’’ means precisely the two norm bounds (1). They hold for every finite coefficient sequence, including sequences placed at any integer translate of the displayed lattice segment. Absolute singular values still scale as $\sqrt{h}$; the theorem controls their ratio, rather than asserting a nonvanishing absolute scale under grid refinement.

3 The physical readout penalty and the sharp-step limit

Define the rectangular difference map $E_m : \mathbb{R}^m \rightarrow \mathcal{W}_m$ by

$$E_m q = (q_0, q_1 - q_0, \dots, q_{m-1} - q_{m-2}, -q_{m-1})^T.$$

It is a bijection onto $\mathcal{W}_m$, and $B_m = P_m E_m$. Both endpoint terms matter.

Proposition 2. *The singular values of E_m , in increasing order, are*

$$e_k = 2 \sin\left(\frac{k\pi}{2(m+1)}\right), \quad 1 \leq k \leq m, \quad \kappa(E_m) = \cot\left(\frac{\pi}{2(m+1)}\right). \quad (5)$$

Moreover,

$$\frac{\kappa(E_m)}{C_\lambda} \leq \kappa(P_m) \leq C_\lambda \kappa(E_m). \quad (6)$$

Thus $\kappa(P_m) = \Theta_\lambda(m)$ at fixed $\lambda > 0$, whereas $\kappa(B_m)$ stays bounded as m increases.

Proof. The matrix $E_m^T E_m$ has diagonal entries 2 and off-diagonal entries -1 . Its sine eigenvectors, with entries $\sin(jk\pi/(m+1))$ for $1 \leq j \leq m$, have eigenvalues $4\sin^2(k\pi/(2(m+1)))$. This proves (5). For $w = E_m q \neq 0$, Theorem 1 gives

$$ha_\lambda \frac{\|q\|_2^2}{\|E_m q\|_2^2} \leq \frac{\|P_m w\|_2^2}{\|w\|_2^2} \leq 4h \frac{\|q\|_2^2}{\|E_m q\|_2^2}.$$

Taking maxima and minima over q yields

$$\frac{\sqrt{ha_\lambda}}{e_1} \leq \sigma_{\max}(P_m) \leq \frac{2\sqrt{h}}{e_1}, \quad \frac{\sqrt{ha_\lambda}}{e_m} \leq \sigma_{\min}(P_m) \leq \frac{2\sqrt{h}}{e_m}.$$

Dividing the respective bounds proves (6). $\square$

This comparison identifies a cumulative-coordinate penalty. It does not assert that every singular value increases under differencing, or that $\kappa(B_m) < \kappa(P_m)$ for every finite pair (m, λ) .

There is an exact limiting comparison. For fixed m, h , as $\lambda \rightarrow \infty$, the densities ρ_λ form an approximate identity. Equation (3) therefore converges in L^2 to $2p_q$, equivalently $\psi_j \rightarrow 2\mathbf{1}_{[jh, (j+1)h]}$ in $L^2(dx)$. In the resulting sharp-step dictionary,

$$B_m^* B_m = 4hI, \quad \kappa(B_m) = 1, \quad \kappa(P_m) = \kappa(E_m) \sim \frac{2(m+1)}{\pi}. \quad (7)$$

The last identity follows from $\|P_m E_m q\|_2^2 = 4h \|q\|_2^2$: the singular values of P_m are $2\sqrt{h}/e_k$, in reverse order. The Gram condition number changes from order m^2 to one. The bound C_λ is conservative and tends to $\pi/2$, so this sharp limit uses the exact argument above.

4 Width uniformity does not remove small-slope attenuation

To see which difficulty remains, periodize the Fourier energy. For every finite q ,

$$\|B_m q\|_2^2 = \frac{h}{2\pi} \int_{-\pi}^{\pi} G_\lambda(\omega) |Q(\omega)|^2 d\omega, \quad G_\lambda(\omega) = 4 \sum_{k \in \mathbb{Z}} \text{sinc}^2\left(\frac{\omega + 2\pi k}{2}\right) M_\lambda(\omega + 2\pi k)^2. \quad (8)$$

This continuous, positive multiplier describes the infinite translated dictionary. Theorem 1 gives $a_\lambda \leq G_\lambda \leq 4$. At the constant and alternating coefficient frequencies,

$$G_\lambda(0) = 4, \quad G_\lambda(\pi) = \frac{8\pi^2}{\lambda^2} \sum_{r=0}^{\infty} \operatorname{csch}^2\left((2r+1)\frac{\pi^2}{2\lambda}\right). \quad (9)$$

Indeed, all nonzero aliases vanish at zero frequency. At π , $\operatorname{sinc}^2(\nu/2) = 4/\nu^2$ for odd multiples ν of π , which cancels the numerator of $M_\lambda(\nu)^2$; pairing positive and negative frequencies gives the series.

Define the infinite-lattice condition number by

$$\kappa_\infty(\lambda) = \sqrt{\frac{\max_{[-\pi, \pi]} G_\lambda}{\min_{[-\pi, \pi]} G_\lambda}}.$$

No assertion about the location of the minimum is needed for

$$\sqrt{\frac{4}{G_\lambda(\pi)}} \leq \kappa_\infty(\lambda) \leq C_\lambda. \quad (10)$$

These modes also have a finite-sequence meaning. The real packets $q_j = m^{-1/2}$ and $q_j = (-1)^j m^{-1/2}$ have Fourier energy tending to point masses at 0 and π , respectively. This is the approximate-identity property of the Fejér kernel $|Q|^2$. Their squared gains tend to $hG_\lambda(0)$ and $hG_\lambda(\pi)$. In particular, $\liminf_{m \rightarrow \infty} \kappa(B_m) \geq \sqrt{4/G_\lambda(\pi)}$. An infinite constant sequence itself is not an ℓ^2 coefficient vector and is not being used as one.

Set $A = \pi^2/(2\lambda)$. As $\lambda \downarrow 0$, the first term in (9) dominates, since $\operatorname{csch}^2((2r+1)A) = 4e^{-2(2r+1)A}(1 + o(1))$. Hence

$$G_\lambda(\pi) \sim \frac{32\pi^2}{\lambda^2} e^{-\pi^2/\lambda}, \quad \sqrt{\frac{4}{G_\lambda(\pi)}} \sim \frac{\lambda}{2\sqrt{2}\pi} e^{\pi^2/(2\lambda)}, \quad C_\lambda \sim \frac{\lambda}{2\pi} e^{\pi^2/(2\lambda)}. \quad (11)$$

Thus the remaining exponential dependence on small λ is real, not merely an artifact of the lower-bound proof. For example, $\lambda = 0.25$ gives $a_\lambda \approx 1.80834 \times 10^{-14}$, $C_\lambda \approx 1.48727 \times 10^7$, and the lower bound in (10) is approximately 1.05166×10^7 . These are whole-line spectral constants, not condition numbers of a particular finite training matrix. The large-width packet limit and the small- λ limit are distinct statements.

5 Finite windows, halo columns, and the missing lower bound

Restricting an integral to $[-1, 1]$ decreases its value. Therefore (1) supplies a finite-window upper bound but does not supply the same lower bound. A bump supported mostly outside the window can have arbitrarily little observed energy. Bias and anchor columns introduce an additional concrete obstruction.

Proposition 3 (Bias-halo cancellation). *Let $\mathbf{B}$ be a real sampled feature matrix with M rows, at least as many rows as columns, and samples $x_i \in [-1, 1]$. Suppose it contains the columns*

$$\frac{d_b}{\sqrt{M}} \mathbf{1}, \quad \frac{t}{\sqrt{M}} \phi_W(x_i), \quad \phi_W(x) = \tanh\left(\frac{\lambda}{h}(x - x_W)\right), \quad x_W = 1 + Rh,$$

where $d_b, t, \lambda, h > 0$ and $R \geq 0$. There is no assumption on its other columns. Then

$$\sigma_{\min}(\mathbf{B}) \leq \frac{2t}{\sqrt{1 + (t/d_b)^2}} e^{-2\lambda R}, \quad \kappa(\mathbf{B}) \geq \frac{\sqrt{d_b^2 + t^2}}{2t} e^{2\lambda R}. \quad (12)$$

The full condition number is infinite if the matrix is rank deficient.

Proof. For every observed x ,

$$0 < 1 + \phi_W(x) = \frac{2e^{2\lambda(x-x_W)/h}}{1 + e^{2\lambda(x-x_W)/h}} \leq 2e^{-2\lambda R}.$$

Choose a coefficient vector v equal to t/d_b in the bias position, 1 in the anchor position, and zero elsewhere. Then $\|v\|_2^2 = 1 + (t/d_b)^2$ and $\|Bv\|_2 \leq 2te^{-2\lambda R}$. This bounds $\sigma_{\min}$ from above. The bias column gives $\sigma_{\max} \geq d_b$, proving the second inequality. $\square$

In particular, $R = \lceil \sqrt{N} \rceil$ gives a lower bound at least $\frac{1}{2}e^{2\lambda\lceil\sqrt{N}\rceil}$ for the full condition number. Increasing λ can improve the interior smoothing spectrum while worsening this out-of-domain cancellation. Neither bound identifies which directions a target residual occupies.

Three further changes require their own estimates. Scaling the bump coefficients by a positive diagonal S gives only $\kappa(B_m S) \leq C_\lambda S_{\max}/S_{\min}$; the scale ratio may depend on width. Replacing integration by samples requires an operator-norm bound on the Gram approximation error to preserve a positive lower bound. Finally, unequal slopes invalidate the common convolution multiplier; same-sign slopes still cancel tails, but opposite signs need not. None of these cases is covered merely by citing Theorem 1.

6 Invertible coordinates and what they control

For a finite window, let $c = (b, w_1, \dots, w_W)^T$ be the physical coefficients and define the complete sampled dictionary by

$$A = M^{-1/2}[\mathbf{1}, \phi_1, \dots, \phi_W].$$

Geometry is frozen here. For any fixed invertible matrix T , set $c = T\theta$. Then

$$\text{range}(AT) = \text{range}(A), \quad \nabla_\theta \mathcal{L} = T^T \nabla_c \mathcal{L}, \quad \Delta c = -\eta T T^T \nabla_c \mathcal{L} \quad (13)$$

for one ordinary gradient-descent step in θ . The first identity preserves approximation capacity. The last identity changes the physical gradient metric, and is exact for any differentiable loss with this fixed linear map. It is not the update rule for Adam.

Here are the concrete maps used in the source implementation review. Let $\alpha_j > 0$ and $\alpha_b > 0$ be fixed reference allowances; they are scales, not constraints imposed on trained coefficients. Put $s_j = \sum_{k=1}^j \alpha_k$. Let C be the square cumulative-sum matrix and $L = C^{-1}$ its lower bidiagonal inverse. In each row of Table 1, $T = \text{diag}(t_b, T_w)$.

Table 1: Invertible maps for the complete physical readout.

Coordinates	Hidden-readout map T_w	Bias scale t_b
Raw	I	1
Collective	$\text{diag}(\sqrt{\alpha_j})$	$\sqrt{\alpha_b}$
Individual	$\text{diag}(\alpha_j)$	α_b
Collective neighboring	$L \text{diag}(\sqrt{s_j})$	$\sqrt{\alpha_b}$
Individual neighboring	$L \text{diag}(s_j)$	α_b

For a neighboring map $w = LS\theta_w$, the feature row is exactly

$$[\phi_1, \dots, \phi_W]LS = [S_{11}(\phi_1 - \phi_2), \dots, S_{W-1,W-1}(\phi_{W-1} - \phi_W), S_{WW}\phi_W].$$

The final anchor makes L square and invertible. This map differs from the rectangular E_m used for the whole-line zero-sum comparison. It induces the coupled physical metric LS^2L^T for hidden readouts. The scales multiply cumulative coefficients before differencing: unequal scales applied to individual tanhs first would leave uncanceled tails in $d_j\phi_j - d_{j+1}\phi_{j+1}$.

A matched physical start must be encoded as $\theta_0 = T^{-1}c_0$. Independent random draws in each coordinate system generally give different physical functions. Likewise, native isotropic damping changes its physical penalty under this map: $\mu \|\Delta\theta\|_2^2 = \mu \|T^{-1}\Delta c\|_2^2$. A claim about coordinates should specify both initialization and the optimizer or regularizer.

Let $y \in \mathbb{R}^M$ contain the target values at the samples. For the squared training loss, $\mathcal{L}(\theta) = \frac{1}{2} \|AT\theta - y/\sqrt{M}\|_2^2$, the Hessian is $(AT)^T(AT)$. With $0 < \eta < 2/\|AT\|_2^2$, an attainable left-singular residual component evolves exactly as

$$e_i(k) = (1 - \eta\sigma_i^2)^k e_i(0), \quad (14)$$

where $\sigma_i > 0$ is the corresponding singular value of AT . Components orthogonal to its range remain unchanged. A tiny positive singular value is therefore a rate obstruction for this algorithm when its residual component is nonzero; it is not an exact-arithmetic error floor. Stability and rates use the complete matrix, including bias, anchor, and halos, not the isolated bound $4h$ from Theorem 1.

These facts locate the role of neighboring readouts in a proposed “ γ barrier.” At fixed spacing, changing γ changes $\lambda = h\gamma$ and hence the dictionary’s smoothing. Changing T at fixed γ preserves its represented function space but changes the gradient metric and its singular spectrum. A frozen comparison at matched physical initialization can therefore separate some coordinate dependence from slope dependence. Reference allowances should be held fixed during that comparison if the intended change is slope alone. The whole-line calculation predicts a small- λ obstruction in fine coefficient modes after removing the cumulative-coordinate penalty. It does not prove that a given target loads those modes, that finite-window halos are harmless, or that a measured fraction of finite-budget optimization error is caused by bounded slopes. Such conclusions need the actual complete dictionary and residual, with the algorithm specified.

Source and scope. This standalone note reorganizes the analytic argument in `docs/neighbor_difference_conditioning.md` and the readout-map audit in `docs/frozen_geometry_capacity_access/pr_conditioning_review.md`. The latter records the maps in the merged implementation at commit `d7d26e6`. The numerical constants above are evaluations of the displayed formulas, as also recorded in the former source. No new experiment or empirical convergence claim is included here.